

PXRD Converter — Instructions

Convert .ras / .brml PXRD files to .xy or Excel, with optional baseline correction and smoothing

What this program does

PXRD Converter is a simple standalone tool for converting powder X-ray diffraction (PXRD) scan files into plain, portable formats. It reads Rigaku (.ras) and Bruker (.brml) files directly — no instrument software required — and exports either a plain .xy coordinate file or a ready-to-use Excel spreadsheet. It also has optional one-click baseline correction and noise smoothing.

Installation

No installation needed. This is a single standalone .exe file — Python does not need to be installed on your computer.

- Copy **PXRD_Converter.exe** to any folder on your computer (Desktop, a lab data folder, a USB drive — anywhere).
- Double-click it to run. A window titled "PXRD File Converter" will open.

Note: Windows SmartScreen may show a "Windows protected your PC" warning the first time you run it, since the .exe isn't digitally signed. This is normal for internally-shared tools. Click **More info** then **Run anyway** to proceed.

How to use it

- **Browse...** — click this and select your scan file (.ras, .brml, or a plain text .xy/.xye/.dat/.txt/.csv/.uxd file). The pattern (intensity vs 2theta) is plotted automatically as soon as it loads.
- **Baseline Correction** (optional) — click to remove the sloping/curved background from the pattern. The plot updates automatically, showing the raw curve, the estimated baseline, and the corrected curve.
- **Smoothing** (optional) — check the box, enter a level (a number like 5, 10, 15, or 25 — similar to Origin's smoothing points setting), then click **Smooth**. Higher numbers smooth more but can start to flatten real peaks, so start low and increase only if needed.
- **Convert to .xy / Convert to Excel (.xlsx)** — exports whatever is currently shown in the preview. If you skipped Baseline Correction and Smoothing, this exports the data exactly as recorded by the instrument, with no processing applied.

Baseline correction and smoothing are both entirely optional. If you just want a plain, unprocessed conversion, skip both steps and go straight to Convert.

Output files

The .xy file contains plain "2theta intensity" coordinates with no headers or comments. The Excel file has two columns labeled "2Theta" and "Intensity (a.u.)", ready to select and paste directly into Origin. If

baseline correction or smoothing was applied, the suggested filename gets a `_corrected` and/or `_smoothed` suffix so it's clear what's in the file.

Supported file formats

- **.ras** — Rigaku (e.g. SmartLab)
- **.brml** — Bruker (DIFFRAC.SUITE)
- **.xy, .xye, .dat, .txt, .csv, .uxd** — plain two-column text files

Note: .raw files are not supported. If your instrument software can export as .ras or .brml instead, use one of those.

Processing methods used

For anyone who wants to know (or cite) exactly what the two optional processing steps do:

Baseline Correction: SNIP algorithm

Method: SNIP — Statistics-sensitive Non-linear Iterative Peak-clipping.

Reference: Ryan, C.G. et al. (1988), "SNIP, a statistics-sensitive background treatment for the quantitative analysis of PIXE spectra in geoscience applications", Nuclear Instruments and Methods in Physics Research B, 34, 396–402.

- The intensity data is passed through a log-log-square-root (LLS) transform, which compresses peak heights relative to the background so peaks don't distort the baseline estimate.
- An iterative clipping window is swept across the transformed data: at each point, the value is replaced with the smaller of itself or the average of its two neighbors at increasing distance. This erodes peaks down toward the background level while leaving flat/background regions untouched.
- The result is transformed back to intensity units, giving a smooth curve following the slowly-varying background, which is then subtracted from the raw pattern.

No manual region selection or peak-picking is required. Implemented directly with numpy (no external XRD-specific library), following the standard SNIP formulation used in PIXE, XRF, and PXRD background subtraction software.

Smoothing: Savitzky-Golay filter

Method: Savitzky-Golay filter (local polynomial smoothing).

Reference: Savitzky, A. and Golay, M.J.E. (1964), "Smoothing and Differentiation of Data by Simplified Least Squares Procedures", Analytical Chemistry, 36 (8), 1627–1639.

- A small window of points slides across the pattern. Within each window, a 3rd-degree polynomial is fit to the data by least squares.
- The fitted polynomial's value at the center of the window replaces the original (noisy) point, and the window moves one step forward.

- Because a local polynomial (not a flat average) is fit, sharp features are preserved far better than with a simple moving average — peak height, position, and width stay close to their true values even as random noise is reduced.

The "level" setting in the app is the window size (number of points), analogous to the smoothing-points setting in Origin. A larger window smooths more aggressively but can start to flatten narrow real peaks — typical values are 5–25 points; start low and increase only if needed.

Implemented directly with numpy (via the standard Savitzky-Golay coefficient derivation, the pseudo-inverse of the local Vandermonde matrix), rather than depending on scipy, to keep the tool's dependencies minimal.

Both steps are entirely optional and only run when their button is clicked. If skipped, the exported file contains the data exactly as read from the instrument file.

Troubleshooting

- **"Windows protected your PC"** — click More info, then Run anyway (see note above).
- **"Conversion failed" / error message on loading a file** — the file may be a format or variant this tool doesn't recognize. Double-check it's a genuine .ras or .brml export, not renamed from something else.
- **The window doesn't open at all** — try running it once from a command prompt (cmd.exe) instead of double-clicking, so any error message stays visible for troubleshooting.

How to cite

If this tool was useful in your work, please cite it via its Zenodo DOI:

Gadekar, C. (2026). *PXRD Converter* (v1.0.0). Zenodo. <https://doi.org/10.5281/zenodo.22896194>

Project page and source code: <https://github.com/Chaitanya09102001/PXRD-Converter-GUI>

Latest release (download the .exe):

<https://github.com/Chaitanya09102001/PXRD-Converter-GUI/releases/latest>